Today

- Simulation based inference

Simulation based inference (SBI)

- Basic idea
- Simulate model with different parameter values
- Compare simulations to data to infer the parameter value of the data
- Back in the 90s, ecologists working on IBMs invented this idea as “pattern oriented modeling”

Bayesian SBI

- General algorithmic idea
  - Draw parameter value from prior
  - Simulate model with that value
  - Does it look like the data?
  - If so, add to set of posterior samples
- Approximate Bayesian Computation (ABC)
- Neural Bayesian Computation

Bayesian SBI

Rejection ABC

Algorithm

for many iterations

Draw parameter θ from prior

Simulate data from model (e.g. IBM)

Compute summary statistic S_{sim}

Compute distance d between S_{sim} and S_{obs}

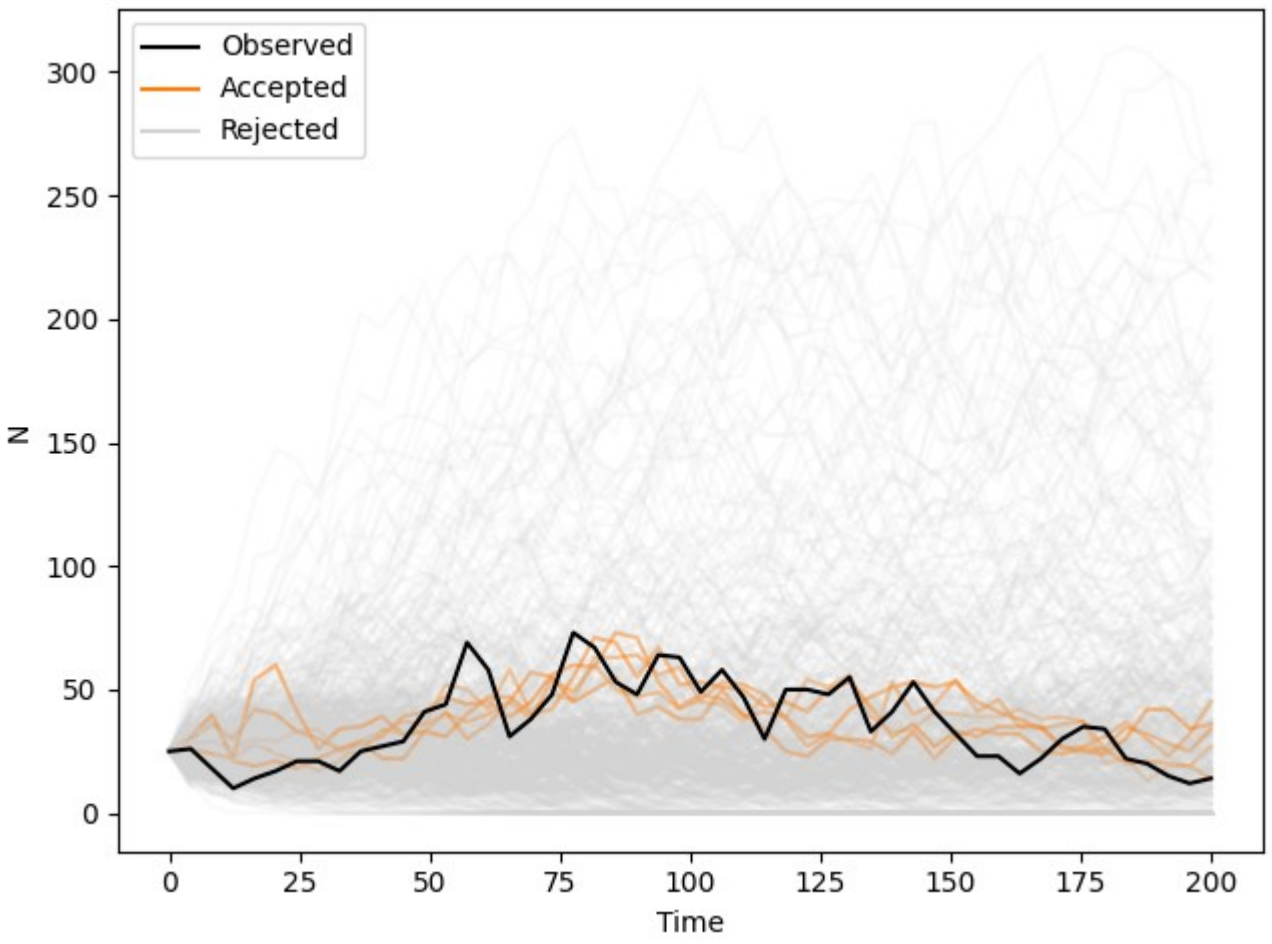
If d < tolerance:

accept θ

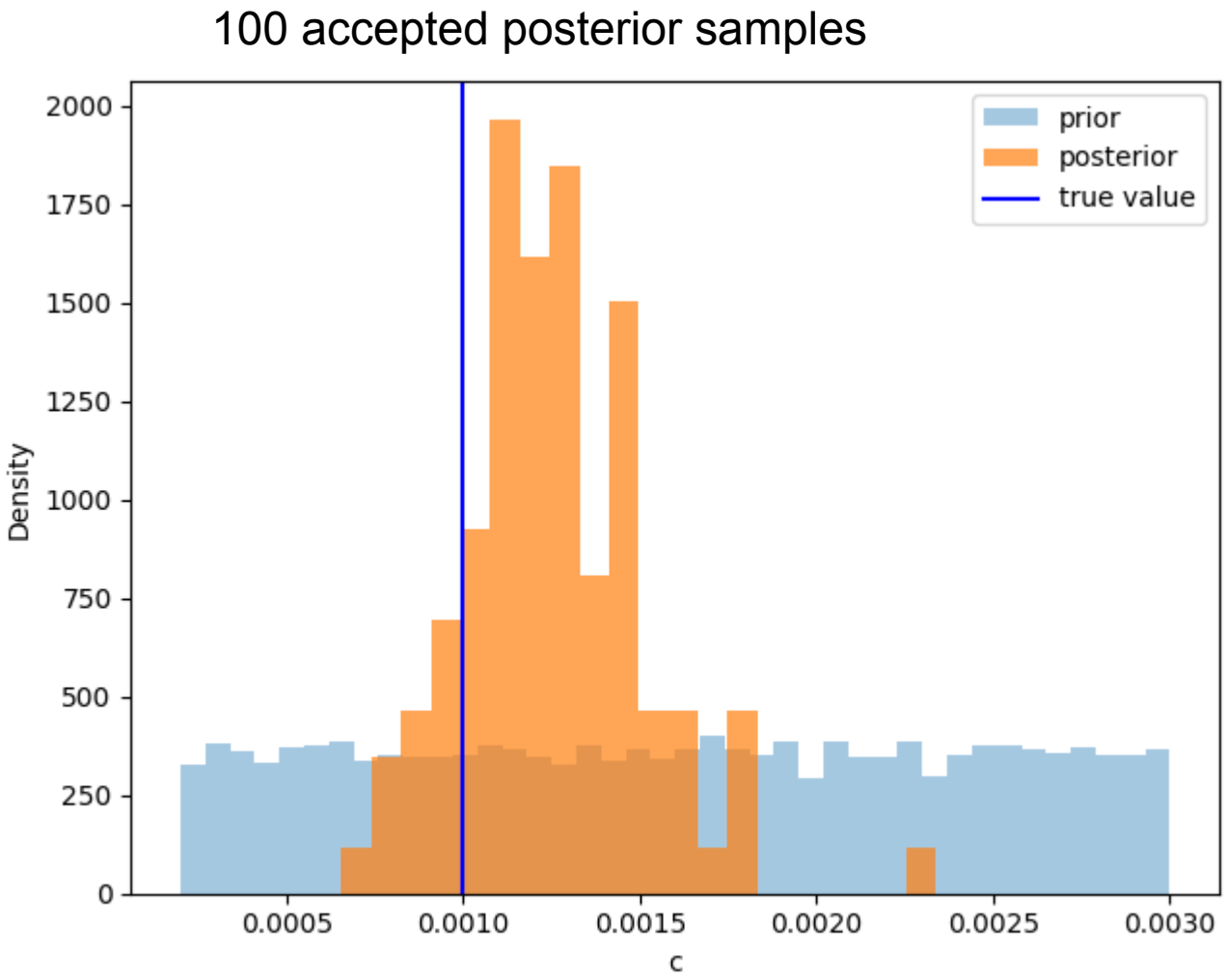
Bayesian SBI

- Summary statistics
  - data itself
  - mean
  - variance
  - max
  - final value

Rejection ABC



Rejection ABC



Bayesian SBI

- Rejection ABC
  - independent proposals for θ
  - wasteful
- ABC-MCMC
  - proposals for θ in high probability regions
  - burn in, convergence, etc
- ABC-SMC
  - sequential Monte Carlo
  - reduce tolerance sequentially
  - “narrowing in”, refining the posterior distribution
  - most modern approaches are variants

Neural SBI

- Train neural network to predict θ given data
- Instead of choosing summary statistics (features), neural network learns which features of the data are informative
- Supervised learning
  - Labeled data
  - pairs: (θ, simulated data)
  - unlimited training data (as many sims as you like)

Neural SBI

Posterior distribution predicted by neural network

